

Rivael Sugianto Sagala

Student at Undergraduate Program in Software Engineering Technology, Institut Teknologi Del

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PROFESSIONAL SUMMARY

Highly motivated last-year Software Engineering Technology student at Institut Teknologi Del with a strong interest in Artificial Intelligence, Machine Learning, and AI-driven systems. Experienced in developing intelligent applications using Python, TensorFlow, and Flask, including deep learning models for image classification with high accuracy. Possesses solid software engineering fundamentals and hands-on experience in building scalable systems using Go and React. Demonstrated research exposure through international training programs, with the ability to analyze problems, experiment with models, and collaborate in structured research environments. Currently seeking an AI Agent Research & Development internship to contribute to innovative AI solutions while strengthening research and technical expertise.

EDUCATION

Del Institute of Technology, Undergraduate Students of Software Engineering Aug 2022 - Sep 2026 (Expected)

- Overall GPA: 3.62/4.0

PROFESSIONAL EXPERIENCE

PT Moonlay Technologies - South Jakarta, Indonesia Aug 2025 – Nov 2025
Software Engineer Intern

- Engineered scalable internal and external enterprise applications by developing dynamic interfaces with React.js, building robust RESTful APIs using Go and Python, and integrating Large Language Models (LLMs) to enhance system intelligence.
- Developed high-quality software delivery within an Agile Scrum environment by collaborating cross-functionally with UI/UX, QA, and senior developers through sprint planning, daily stand-ups, and rigorous code reviews.

University of Chicago - South Ellis Avenue, Chicago, USA Jan 2025 – Jun 2025
Student, Indonesia Research Training Program

- Selected to participate in a prestigious international research training program led by Prof. Haryadi Gunawi from the University of Chicago.
- Underwent comprehensive training focused on core research competencies, including identifying impactful research problems, critically analyzing academic literature, engaging with the right research communities, reproducing scholarly work, and conducting research at a top-tier level.

Institut Teknologi Del - Toba, North Sumatera, Indonesia Jan 2025 - May 2025
Full Stack Developer

- Developed a village financial data platform to enhance budget management and reporting for over 411 households.
- Utilized Flowbite UI components and Tailwind CSS for front-end development and integrated backend features using Laravel and MySQL.
- Improved budget management and reporting efficiency by 90%, ensuring better accessibility and accuracy in financial data.

PROJECT EXPERIENCE

Cognitive Operation Receptionist Agent (CORA)

Full Stack Developer

- Developed an AI-powered virtual receptionist system integrating Computer Vision, Speech-to-Text, Text-to-Speech, and LLM-based conversation handling.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline for knowledge-based question answering using document embeddings and contextual memory.
- Built frontend features using React & TypeScript, and backend services using Python (Flask) following clean architecture principles.

Electronic Capability Study Report (E-CSR)

Frontend Developer Intern

- Developed an enterprise-scale web application to digitize capability study reports for the aviation industry's

Maintenance, Repair, and Overhaul (MRO) sector.

- Built a responsive and modular frontend architecture using React.js and TypeScript, ensuring type safety and implementing complex state management for dynamic data forms.
- Collaborated within an Agile Scrum team, actively participating in Sprint Planning, Daily Stand-ups, and rigorous Code Reviews via GitHub Pull Requests.

Fish Species Classification from Lake Toba using CNN and ANN

Full Stack Developer

- Built a responsive web app using HTML, CSS, Bootstrap (front end) and Flask with Python (back end) for classifying Lake Toba fish species using CNN and ANN, achieving 97% accuracy.
- Integrated image upload and real-time detection using OpenCV for enhanced user interaction.

Puzzle Game with Image Processing and Backtracking Algorithm

Full Stack Developer

- Developed a web-based puzzle game using image processing and backtracking algorithm with a 95% success rate, built with Python and Flask.
- Designed a responsive interface using HTML, CSS, JavaScript, and Bootstrap.

ORGANIZATION EXPERIENCE

Communication and Information Department, Creative division- BEM

Jan 2024 - Present

Member

- Designed visual content for social media and campus promotions using consistent branding strategies.
- Collaborated on creative strategies and documented key organizational activities.

Cyber Security - UKM

Oct 2023 - Oct 2024

Member

- Participated in cybersecurity workshops and collaborated on team projects related to network security.
- Helped organize knowledge-sharing sessions and internal events to promote cybersecurity awareness.

Communication and Information Department, Information System Division - BEM

Sep 2023 - Aug 2024

Member

- Maintained and updated digital platforms for internal and external communications.
- Supported the development and management of information systems to improve communication efficiency.

COMPETITIONS AND CERTIFICATIONS

Huawei ICT Competition 2024 – **Competition**, Huawei

Mobile Programmer Vocational School Graduate Academy – Competition, Digital Talent Scholarship by Kominfo

Laravel Web Development – **Certification**, SanberCode

Flutter & Dart: The Complete Guide – **Certification**, Udemy

Intro to Data Analytics – **Certification**, RevoU

Introduction to Software Engineering – **Certification**, IBM

Supervised Machine Learning: Regression and Classification – **Certification**, DeepLearning.AI

TECHNICAL SKILLS

Programming Languages: Python, C, Java, JavaScript, Go, PHP, Kotlin, Dart

Frameworks: React, Laravel, Node.js, React.js, React Native

Databases: MySQL, Firebase, SQLite, MongoDB, Cassandra

Libraries: Scikit-learn, Pandas, Flask, Matplotlib, TensorFlow, Keras, Seaborn

Tools & Miscellaneous : Overleaf, GitHub, Visual Studio Code, Figma, Trello, Tableau, Arduino IDE

LANGUAGES

Indonesia - Full Professional Proficiency

English - Conversational English Proficiency